

Theory

A **Fuzzy Logic Controller (FLC)** is a rule-based system that mimics human reasoning for decision making under uncertainty. Instead of using precise numerical inputs, it uses **linguistic variables** (e.g., *poor*, *average*, *good*) and applies **fuzzy rules** to produce smooth, human-like control actions.

1. Objective

To design a **Fuzzy Tipping Controller** that determines the **tip percentage** based on two inputs:

- **Service Quality** (0–10)
- **Food Quality** (0–10)

and outputs:

- **Tip (%)** (0–25)
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2. Fuzzy Variables and Membership Functions

Variable	Type	Linguistic Terms	Range
Service	Input	Poor, Average, Good	0–10
Quality	Input	Poor, Average, Good	0–10
Tip	Output	Low, Medium, High	0–25

Membership functions are **triangular**, representing gradual transitions between terms.

3. Rule Base (Mamdani Type)

1. IF *Service* is **Poor** OR *Quality* is **Poor**, THEN *Tip* is **Low**
2. IF *Service* is **Average**, THEN *Tip* is **Medium**
3. IF *Service* is **Good** OR *Quality* is **Good**, THEN *Tip* is **High**

These rules form a **Mamdani inference system**, where both antecedent and consequent are fuzzy sets.

4. Working Principle

1. **Fuzzification:** Converts crisp inputs (quality, service) into fuzzy degrees using membership functions.
 2. **Rule Evaluation:** Applies the defined IF–THEN rules using fuzzy logic operations (AND, OR, MIN, MAX).
 3. **Aggregation:** Combines outputs from all rules into a single fuzzy set for the “tip.”
 4. **Defuzzification:** Converts the fuzzy result back into a crisp numeric tip value (using the **centroid** method).
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5. Output Behavior

- When both *service* and *quality* are poor → **Low Tip** (~0–5%)
- When both are average → **Medium Tip** (~10–15%)
- When both are good → **High Tip** (~20–25%)

The controller smoothly transitions between these levels based on fuzzy reasoning.

6. Advantages

- Handles vague or imprecise input data.
- Simple, rule-based design—no mathematical model needed.
- Produces smooth, human-like control decisions.

7. Disadvantages

- Rule base must be carefully defined.
 - Performance depends on shape and number of membership functions.
 - Not ideal for high-dimensional or rapidly changing systems.
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Conclusion

The **Fuzzy Tipping Controller** demonstrates how **Mamdani Fuzzy Inference** can model real-life decision making. By combining linguistic rules and fuzzy logic, it provides a logical yet flexible way to compute tips based on subjective inputs like service and food quality.